

GLOBAL MACRO TRADING & RISK ENGINE

Goldman Sachs FICC & Equities Portfolio Research Project

Full plain-English guide, methodology, models, results and interview preparation | 2000-2025 back-test | 2026 market overlay

The project in one sentence

I built a repeatable framework that uses economic and market information to decide how much risk to take, then tested the idea over 26 years, measured both returns and risk, and added specialist models for rates, FX, options, credit, factors, optimisation and stress testing.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
GLOBAL MACRO TRADING & RISK ENGINE V4													
Investment-research style project 2000-2025 backtest + 2026 market overlay Educational only, not investment advice													
HEADLINE RESULTS					CURRENT OVERLAY								
Metric	Strategy	S&P 500	60/40	Equal-weight 7	Indicator	Value	Regime link	Source					
Cumulative return	461.1%	638.8%	472.9%	493.8%	S&P 500	7591.7 2026 YTD +10.9%	https://fred.stlouisfed.org/series/SP500						
CAGR	6.9%	8.0%	6.9%	7.1%	Fed Funds	3.6% Below 2025 avg 4.21%	https://fred.stlouisfed.org/series/FEDFUNDS						
Volatility	6.2%	17.8%	9.9%	7.4%	US 10Y yield	4.7% Rates context	https://fred.stlouisfed.org/series/DGS10						
Max drawdown	(11.0%)	(36.6%)	(18.0%)	(10.1%)	Brent	99.09 Above 2025 avg \$69.14	https://fred.stlouisfed.org/series/WOBLBRENTU						
					MODEL STACK	Why it matters							
INTERVIEW-WORTHY FINDINGS													
Downside lesson	The adaptive strategy prioritises diversification and risk control rather than simply maximising equity exposure.	Real Estate reduces the portfolio's dependence on the traditional stock/bond/gold trio.											
					Real Estate sleeve	Adds another diversifying return stream to the portfolio							
Portfolio lesson					Risk parity	Tests weights based on inverse volatility							
Rates lesson		Duration and DV01 show how a bond portfolio reacts when yields move by basis points.			Min variance	Finds a lower-volatility long-only allocation							
FX lesson		GBP/USD carry can be decomposed into rate differential + spot movement; spot can dominate small carry.			Max Sharpe	Finds the best historical excess-return / volatility trade-off							
Derivatives lesson		Options Greeks explain how price, volatility and time affect an option before any trade is made.			Monte Carlo	Shows distribution of possible terminal wealth under assumptions							
Research mindset		The model documents where assumptions fail and what the next version should improve.			Options + Rates	Black-Scholes Greeks + duration/convexity/ DV01							
CURRENT GROWTH REGIME ALLOCATION													
S&P 500	30.0% Small Cap		10.0% 10Y		15.0% Baa		10.0% Real Estate		10.0% Gold		10.0% Cash		15.0%
Composite score	1 Regime		Growth										

Figure 1. Main dashboard from the executed model.

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1. Executive summary

This project is designed as a self-directed introduction to institutional-style global markets research. It is deliberately broader than a stock-picking exercise. The central question is whether macroeconomic and market information can be converted into a repeatable risk-allocation process, and whether that process can reduce drawdowns without giving up all long-run return.

- Historical sample: 1998-2025, with the formal back-test beginning in 2000; 1998-1999 are warm-up observations.
- Back-test frequency: annual total-return data, so the exercise is a research model rather than a live-trading simulation.
- Portfolio: seven investment sleeves plus cash - US large-cap equities, US small-cap equities, US 10Y Treasury bonds, Baa corporate bonds, US real estate, gold and cash.
- Core signals: monetary-policy/rates, energy-price/inflation proxy, equity momentum and a credit-cycle signal.
- Model stack: regime engine, back-test, attribution, portfolio optimisation, risk analytics, performance diagnostics, Monte Carlo, bond analytics, Black-Scholes options, FX carry/scenarios, factor analysis, credit model and scenario stress tests.

Metric	Strategy	S&P 500	60/40	Equal-weight 7
Cumulative return	448.0%	638.8%	472.9%	493.8%
CAGR	6.8%	8.0%	6.9%	7.1%
Volatility	6.9%	17.8%	9.9%	7.4%
Maximum drawdown	-11.0%	-36.5%	-18.0%	-10.1%
Sharpe	0.73	0.43	0.55	0.73

Main conclusion

The strategy did not beat the S&P 500 on raw return. Its historical value proposition was different: lower volatility and materially smaller drawdowns. That makes the project more interesting than a cherry-picked “winning” back-test because it creates a real trade-off to discuss: capital preservation versus upside capture.

2. The whole idea in very simple English

Imagine you have £100 and you can put it into different buckets. Some buckets are risky but can grow quickly; others are steadier and may protect you when markets fall. The model tries to answer one question: “Given what I currently know about the economy and markets, how much of my £100 should be in each bucket?”

Step	What it means
1. Collect clues	Look at interest rates, oil prices, equity momentum and credit-market behaviour.
2. Turn clues into signals	Each clue becomes +1, 0 or -1. Positive means supportive; negative means caution.
3. Add the signals	The combined score creates a simple market regime such as Risk-on, Growth, Balanced or Defensive.
4. Choose a portfolio	Each regime maps to different weights across equities, bonds, credit, real estate, gold and cash.
5. Test the idea	Use only information that would have been available at the time, then see what the portfolio would have returned.

6. Measure risk	Do not stop at return. Measure volatility, drawdown, beta, correlation, VaR and tail loss.
7. Try to break it	Run stress scenarios such as a stock-market crash, inflation shock or rates shock.
8. Learn from failure	If the model avoids a fall but misses a rebound, that tells you something important about the model.
The key mindset The aim is not to build a machine that “knows” the future. It is to build a transparent process for making decisions when the future is uncertain.	

3. Data and assumptions

The workbook uses annual historical return series and macro inputs, with 1998-1999 acting as warm-up years so the first formal portfolio decisions in 2000 can use lagged information. The model audit explicitly checks that the back-test uses prior-year and prior-two-year inputs rather than future observations.

Data / source	Used for	Why it matters
Damodaran historical returns	S&P 500, small cap, 10Y bonds, Baa corporate bonds, real estate, gold, T-bills	Long historical asset-return series
FRED / Fed Funds	Monetary-policy signal	Tracks the rate environment
FRED / EIA Brent	Energy/inflation proxy	Captures major changes in energy prices
FRED current series	Current market overlay	Provides a live-ish context layer, separate from the historical back-test
Bank of England	UK Bank Rate	Used in the FX context and current overlay
Important limitation Annual data is coarse. A real trading desk would normally work with daily or intraday total-return and market data, transaction costs, financing, liquidity and more precise event timing. The project is valuable as a research exercise because the logic is visible and explainable, not because annual back-testing proves future profitability.		

4. The macro regime engine

The regime engine is the heart of the project. It translates several market/economic clues into a single label that drives portfolio allocation.

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INTERVIEW-WORTHY FINDINGS					MODEL STACK		Why it matters						
Downside lesson	The adaptive strategy prioritises diversification and risk control rather than simply maximising equity exposure. Real Estate reduces the portfolio's dependence on the traditional stock/bond/gold trio. Duration and DVO1 show how a bond portfolio reacts when yields move by basis points. GBP/USD carry can be decomposed into rate differential + spot movement; spot can dominate small carry. Options Greeks explain how price, volatility and time affect an option before any trade is made. The model documents where assumptions fail and what the next version should improve.				Real Estate sleeve	Adds another diversifying return stream to the portfolio							
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Composite score	1 Regime		Growth										

The dashboard summarises the model stack and current growth-regime allocation.

Signal	Simple rule / idea	What the model is trying to detect
Rate signal	Change in Fed Funds rate over prior observations	Tightening vs easing monetary conditions
Energy signal	Change in Brent crude price	Inflation / macro stress proxy
Momentum	Prior S&P 500 total return	Whether risk appetite has positive or negative trend
Credit	Baa corporate bonds relative to government bonds	Credit-cycle / risk appetite information

Once the signals are combined, the model chooses one of four broad regimes. Think of a regime as a weather forecast for markets: it is not the weather itself, but a compact description that helps you choose how prepared you want to be.

Regime	Plain-English meaning	Portfolio idea
Risk-on	Evidence broadly supports taking more risk	Higher equity / growth exposure
Growth	Growth conditions look constructive but not extreme	Meaningful risk assets plus diversification
Balanced	Evidence is mixed	Middle-ground allocation
Defensive	Evidence points to greater downside risk	More bonds, gold and cash; less equity risk

5. Portfolio construction

The expanded project uses seven investment sleeves plus cash. The point of adding more sleeves is to make the model less dependent on one classic stock/bond/gold combination and to demonstrate that different assets respond to different economic drivers.

Sleeve	What it is	Why it is included
S&P 500	Large US equities	Core growth / risk asset
US small cap	Smaller US companies	Higher growth sensitivity / higher beta
US 10Y Treasury	Government bonds	Duration and defensive diversification
Baa corporate bonds	Investment-grade corporate credit at the Baa level	Carry plus credit-cycle exposure
US real estate	Real-estate return proxy	Alternative real-asset diversification
Gold	Commodity / store-of-value proxy	Potential diversifier in inflation or risk-off environments
Cash	Short-term risk-free proxy	Liquidity and drawdown control

Why add real estate?

Real estate creates a separate economic exposure from the traditional stock/bond/gold trio. It is not a perfect hedge - its sensitivity to interest rates and property conditions means it can fall at the same time as equities - but it adds another return stream.

PORTFOLIO MODEL LAB RISK PARITY + MIN VARIANCE + MAX SHARPE + EDITABLE MIX																
Asset	S&P 500	US Small Cap	US 10Y T Bond	Baa Corp Bond	Real Estate	Gold										
Historical mean	9.6%	11.9%	4.2%	6.6%	4.9%	12.3%										
Historical volatility	17.8%	26.9%	8.8%	7.3%	6.7%	17.9%										
Risk parity weight	10.2%	6.7%	19.9%	25.4%	27.6%	10.2%										
Min variance weight	-	0.0%	22.0%	15.8%	54.2%	7.0%										
Max Sharpe weight	6.4%	0.0%	14.7%	13.9%	48.9%	16.1%										
Current model weight	30.0%	10.0%	15.0%	10.0%	10.0%	10.0%										
EDITABLE PORTFOLIO LAB blue cells are user assumptions																
Metric	S&P 500	US Small Cap	US 10Y T Bond	Baa Corp Bond	Real Estate	Gold										
Risky asset weights	30.0%	10.0%	15.0%	10.0%	10.0%	10.0%										
Cash weight	15.0%															
Weight check	100.0%															
Expected return	7.4%															
Portfolio volatility	8.0%															
Sharpe ratio	68.0%															
Rebalance note																
REFERENCE MODEL	Expected return	Volatility	Sharpe	Max drawdown	Comment	S&P 500	US Small Cap	US 10Y T Bond	Baa Corp Bond	Real Estate	Gold					
Risk parity	6.9%	5.6%	84.5%		Inverse-vol weighting across six risky assets	0.0310	0.0300	-0.0050	0.0047	0.003414027	0.003303253					
Minimum variance	5.5%	4.0%	83.8%		Optimised variance subject to long-only weights	-0.0050	-0.0099	0.0081	0.0025	-0.001720484	0.00349311					
Maximum Sharpe	6.5%	4.6%	95.2%		Optimised excess-return / volatility trade-off	0.0047	0.0078	0.0025	0.0050	-0.000135585	0.004628991					
Current editable mix	7.4%	8.0%	68.0%		Change blue weights + cash weight	0.0034	0.0062	-0.0017	-0.0001	0.004235473	-0.00384522					
						0.003303253	0.005320602	0.00349311	0.004628991	-0.00384522	0.030818261					
What it demonstrates	Diversification is about covariance, not just counting assets. A low-return asset can improve the whole portfolio if it reduces risk. Optimisation outputs are historical reference results computed from the 2000-2025 sample; they are not forecasts. Dandorian historical returns, 2000-2025; covariance is sample covariance.															
Important caveat																
Data source																

Figure 2. Portfolio model lab: comparing risk-parity, minimum-variance and maximum-Sharpe constructions.

6. 26-year back-test and what it means

The formal back-test covers 2000-2025. The comparison is deliberately broad: the adaptive regime strategy is set against the S&P 500, a classic 60/40 stock-bond mix and a simple equal-weight seven-sleeve portfolio.

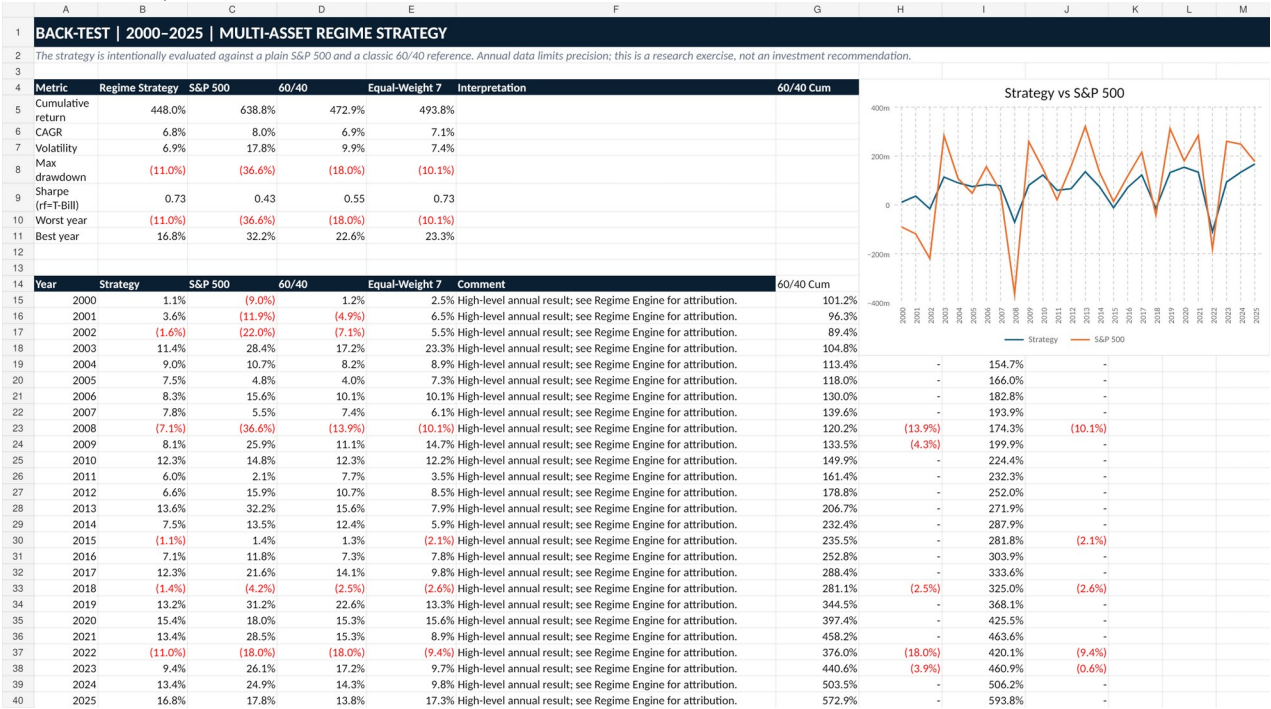


Figure 3. Back-test output from the expanded model.

How to read the result

The strategy generated a lower raw return than the S&P 500, but it also showed substantially lower volatility and drawdown. In other words, the model is better described as a risk-control strategy than as a market-beating strategy.

Year	Strategy	S&P 500	Comment
2000	1.1%	-9.0%	High-level annual result; see Regime Engine for attribution.
2001	3.6%	-11.8%	High-level annual result; see Regime Engine for attribution.
2002	-1.6%	-22.0%	High-level annual result; see Regime Engine for attribution.
2008	-7.1%	-36.5%	High-level annual result; see Regime Engine for attribution.
2009	8.1%	25.9%	High-level annual result; see Regime Engine for attribution.
2013	13.6%	32.1%	High-level annual result; see Regime Engine for attribution.
2020	15.4%	18.0%	High-level annual result; see Regime Engine for attribution.
2022	-11.0%	-18.0%	High-level annual result; see Regime Engine for attribution.
2023	9.4%	26.1%	High-level annual result; see Regime Engine for attribution.
2024	13.4%	24.9%	High-level annual result; see Regime Engine for attribution.
2025	16.8%	17.8%	High-level annual result; see Regime Engine for attribution.

The most useful discussion point is not a single winning year. It is the shape of the trade-off across different market environments: defensive positioning can soften losses, but the same positioning can leave money on the table when markets rebound strongly.

7. Performance attribution and diagnostics

Performance attribution answers: “Where did the return come from?” Instead of looking only at the portfolio total, the model decomposes the contribution of equities, small caps, government bonds, corporate bonds, real estate, gold and cash.

Diagnostic	Value	Plain-English meaning
Tracking error	12.9%	How differently the strategy behaves from the S&P 500
Information ratio	-0.19	Active return per unit of benchmark-relative risk
Correlation vs S&P	0.83	How similarly the strategy tends to move
Beta vs S&P	0.30	Sensitivity to S&P moves
R-squared	0.69	Share of variation explained by the simple benchmark relationship
Upside capture	56.0%	Share of positive benchmark moves captured
Downside capture	11.1%	Share of negative benchmark moves participated in
Same-direction years	92.3%	Frequency with which strategy and benchmark have the same sign

Why this is interesting

The low beta and low downside capture support the idea that the strategy is meaningfully different from a pure equity portfolio. The negative information ratio is also useful: the strategy underperformed the benchmark, so there is no case for calling it superior on every dimension.

8. Risk analytics

Metric	Value	What it means
Expected return	7.4%	Historical mean return of the current mix
Volatility	8.2%	How jumpy annual returns have been
Sharpe ratio	0.66	Return relative to volatility and cash
5% historical VaR	-8.9%	A rough 5th-percentile annual downside estimate
5% Expected Shortfall	-12.1%	Average outcome inside the worst 5% tail
Worst historical year	-13.3%	Largest annual loss observed in the sample

Risk contribution. The workbook also asks which sleeves are actually responsible for portfolio risk. This matters because a 10% position in a very volatile asset can contribute more risk than a larger position in a stable asset.

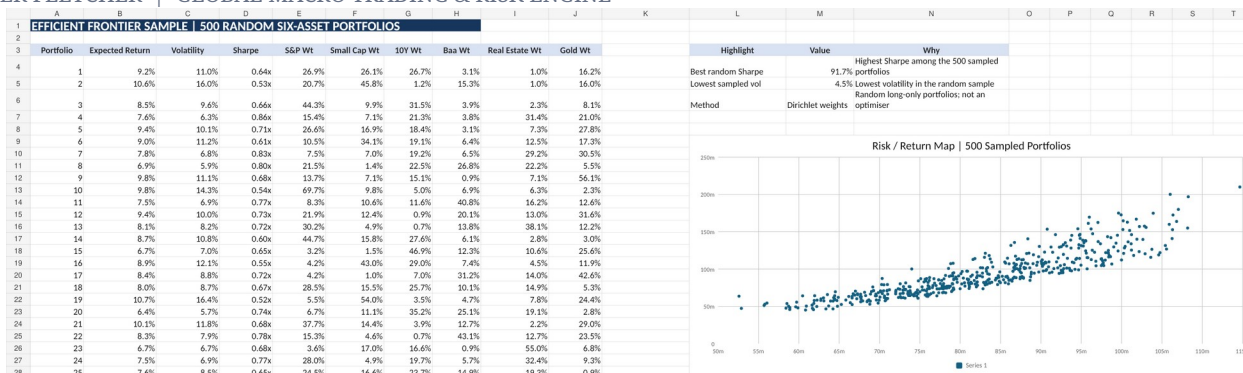


Figure 4. Efficient-frontier style comparison used in the optimisation lab.

Dummy explanation of diversification

Diversification does not mean “own lots of things”. It means owning things that do not all lose money for the same reason at the same time. Correlation is one way of checking whether that is actually happening.

9. The extra investment models

9.1 Portfolio optimisation

The optimisation lab compares three classic constructions: risk parity, minimum variance and maximum Sharpe. It also generates a large cloud of random portfolios to visualise the efficient frontier.

Model	Dummy explanation	Strength	Weakness
Risk parity	Try to spread risk rather than money evenly	Reduces dominance by the most volatile assets	Can produce large weights in low-volatility assets
Minimum variance	Find the mix with the lowest historical volatility	Strong risk-control focus	Can become defensive and concentrated
Maximum Sharpe	Find the best historical excess-return / volatility trade-off	Useful benchmark for risk-adjusted efficiency	Highly sensitive to historical estimates
Efficient frontier	Show many possible return/risk combinations	Makes portfolio trade-offs visual	Historical frontier is not a forecast

9.2 Fixed income: duration, DV01 and convexity

A bond price generally moves opposite to its yield. Duration gives a first approximation of how sensitive price is to yield changes.

Simple formula

Approximate % price change $\approx$ -Duration x change in yield + 0.5 x Convexity x (change in yield)²

Illustrative input	Value
Base bond price	\$100
Yield	4.67%
Modified duration	8.0
Convexity	70
Notional	\$10m

For example, a +100 basis point yield shock produces a large negative price move. This is useful in an interview because it demonstrates that “rates risk” can be expressed quantitatively, not just described qualitatively.

9.3 Options: Black-Scholes and Greeks

The model contains an illustrative equity-index option pricer. It is not a live quote; it is a learning tool for understanding how option value responds to spot, strike, time, interest rates, dividends and volatility.

Input / Greek		Value / example				
Spot		7591.7				
Strike		7591.7				
Expiry		1 year				
Risk-free rate		3.63%				
Dividend yield		1.20%				
Volatility		18%				
Call value		626.33				
Put value		446.25				
Call delta		0.582				
Gamma		0.000281				
Vega per +1 vol point		29.18				
Call theta per day		-0.95				

	A	B	C	D	E	F
1						
2						
3	Input / output	Value	Units	Comment	Source	Type
4	Spot S	7591.70	index	Current S&P 500 level	https://fred.stlouisfed.org/series/SP500	Observed
5	Strike K	7591.70	index	At-the-money	-	Assumption
6	Time to expiry T	1.00	years	One year	-	Assumption
7	Risk-free r	0.04	%	Fed Funds proxy	https://fred.stlouisfed.org/series/FF	Observed proxy
8	Volatility sigma	0.18	%	Illustrative implied/realised volatility assumption	-	Assumption
9	Dividend yield q	0.01	%	Illustrative dividend yield assumption	-	Assumption
10	d1	0.22		Black-Scholes intermediate	-	Calculated
11	d2	0.04		Black-Scholes intermediate	-	Calculated
12	Call value	626.33	index points	Theoretical European call	-	Calculated
13	Put value	446.25	index points	Theoretical European put	-	Calculated
14						
15	Greek	Call	Put	Units	Interpretation	
16	Delta	0.5820	(0.4061)	per 1 index point	Directional sensitivity	
17	Gamma	0.0003	0.0003	per point^2	Delta changes as spot moves	
18	Vega	29.1773	29.1773	index points per +1 vol pt	Volatility sensitivity	
19	Theta	(347.2229)	(171.4821)	points / year	Time decay; convention varies	

Figure 5. Illustrative Black-Scholes output and Greeks.

9.4 FX: GBP/USD carry and scenarios

The FX model starts from GBP/USD and the UK-US interest-rate differential. A simple carry proxy is the rate differential; then a scenario grid shows how spot changes and differential changes could affect a GBP/USD position.

Input	Current snapshot
GBP/USD spot	1.3528
UK Bank Rate	3.75%
US Fed Funds	3.63%
UK-US differential	0.12%
1Y carry proxy	0.12%

FX MODEL GBP/USD CARRY, RATE DIFFERENTIAL & SCENARIO GRID						
Input	Value	Units	Source	Type	Why it matters	
GBP/USD spot	1.3528 USD per GBP		https://www.exchangerates.org.uk/Pounds-to-Dollars-currency-conve	Observed	Starting FX level	
UK Bank Rate	3.75% %		https://www.bankofengland.co.uk/monetary-policy/interest-rates-and	Observed	UK short rate proxy	
US Fed Funds	3.63% %		https://fred.stlouisfed.org/series/FF	Observed	US short rate proxy	
Rate differential (UK-US)	0.12% %	-		Calculated	Simple carry proxy	
1Y long-GBP carry proxy	0.12% %	-		Calculated	Approx. interest differential before spot/basis effects	
Carry + 5% spot shock	5.12% %	-		Scenario	Shows carry plus FX move	
Spot shock	-10%	-5%	0%	+5%	+10%	Illustrative total return on GBP held vs USD
-2% rate diff shock	(11.9%)	(6.9%)		(1.9%)	3.1%	8.1%
-1% rate diff shock	(10.9%)	(5.9%)		(0.9%)	4.1%	9.1%
+0% rate diff shock	(9.9%)	(4.9%)		0.1%	5.1%	10.1%
+1% rate diff shock	(8.9%)	(3.9%)		1.1%	6.1%	11.1%
+2% rate diff shock	(7.9%)	(2.9%)		2.1%	7.1%	12.1%
Interpretation						
Positive rate differential	Higher UK short rates than US rates can support GBP carry, all else equal.					
Spot dominates	A 5% currency move can easily outweigh a small rate differential over one year.					
FICC relevance	FX combines macro expectations, rates and relative value; the model makes those links explicit. This is a simplified carry proxy; real FX pricing incorporates forwards, cross-currency basis, funding, convexity and expectations.					
Caveat						

Figure 6. FX model scenario grid.

9.5 Factor model

The factor model asks how each asset behaves relative to the S&P 500. Beta tells you sensitivity; correlation tells you how closely returns move together; alpha in this workbook is a simple historical intercept and should not be treated as a forward forecast.

Asset	Beta vs S&P	Correlation	Historical alpha
US Small Cap	1.05	0.69	1.93%
US 10Y Treasury	-0.18	-0.36	5.91%
Baa Corporate Bond	0.16	0.40	4.99%
Gold	0.16	0.16	10.82%

9.6 Credit model

The credit model compares Baa corporate-bond returns with the government-bond proxy. When corporates outperform, the signal is treated as a more supportive credit environment; when they lag, the model becomes more cautious. A real credit desk would use spreads, default probabilities and more detailed credit curves.

10. Stress testing

Stress testing asks “what if?” rather than “what is most likely?” The model lets you shock each asset sleeve and see how the current portfolio would react.

Scenario	Typical idea
Risk-off	Equities and credit fall; Treasuries and gold help
Inflation shock	Equities/bonds fall while gold and energy may rise
Rates shock	Yields jump and duration assets fall
Credit shock	Corporate bonds and risk assets weaken
Gold spike	Gold jumps as a diversifier; risk assets fall
Severe stagflation	Large equity/credit losses, higher gold, mixed rates response
S&P crash only	Equities fall sharply while other sleeves cushion
Rates + inflation	Traditional balanced portfolios can face pressure on both

Dummy explanation

A stress test is like dropping your phone on purpose to see whether the case works. You are not predicting the drop; you are checking whether the system survives a bad event.

11. Current market overlay

The historical back-test and the current overlay are intentionally separated. The historical model asks what happened in the past; the overlay asks what the project would say today using the same framework.

Metric	Value	As of
S&P 500	7591.7	2026-09-11
Fed Funds	3.63%	Aug 2026
US 10Y	4.67%	2026-09-10
UK Bank Rate	3.75%	Sep 2026
GBP/USD	1.3528	2026-09-11
Brent annual avg 2025	\$69.14	2025

Current model output

Current regime: Growth. Current mix: 30% S&P 500 / 10% Small Cap / 15% US 10Y / 10% Baa / 10% Real Estate / 10% Gold / 15% Cash. The workbook describes the stance as balanced risk-taking with material liquidity and identifies an upside inflation / rates shock as a primary risk.

12. What I would improve next

- Replace annual return proxies with daily total-return data and exact event timestamps.
- Use actual inflation surprises, central-bank expectations and futures-implied rate paths rather than only realised policy rates.
- Model bid/ask spreads, slippage, financing and turnover so that implementation costs are more realistic.
- Add rolling walk-forward optimisation so the model parameters are re-estimated using only the information then available.
- Use a richer credit model with corporate spreads, default rates and sector breakdowns.
- Add volatility targeting and position-size limits so risk is explicitly budgeted.
- Test the model across different start/end dates to reduce dependence on one sample period.
- Separate research code, production-style calculations and visualisation so the model is easier to audit and reproduce.

The sophisticated point

The next version should be less about adding more indicators and more about making the existing logic harder to fool. Better out-of-sample testing, cleaner data and realistic implementation costs are more valuable than simply adding dozens of signals.

13. What Goldman should notice

Curiosity	Went beyond a personal stock portfolio to study macro, FX, rates, credit and derivatives
Analytical thinking	Built signals, back-tests, attribution and factor/risk diagnostics
Risk awareness	VaR, expected shortfall, drawdown, stress testing and risk contribution
Commercial judgement	Turns research into a portfolio and a hypothetical trade book
Self-starting behaviour	Entire project designed and executed independently
Intellectual honesty	Records benchmark underperformance and model weaknesses
Communication	The framework is designed to be explained from a simple question to a portfolio decision

14. Interview-ready explanation

30-second answer

I wanted to understand markets beyond simply following share prices, so I built a global macro trading and risk engine. It uses rates, energy prices, equity momentum and credit conditions to classify market regimes, then changes the allocation across equities, bonds, credit, real estate, gold and cash. I back-tested it over 2000-2025 and found that it materially reduced volatility and drawdown versus the S&P, although it gave up some upside. I then added optimisation, VaR, Monte Carlo, bond duration, options Greeks, FX scenarios and stress tests to understand the portfolio from different angles.

Best “what did you learn?” answer

The biggest lesson was that reducing downside is not free. A defensive model can protect capital during a sell-off but can also miss a fast rebound. That made me think less about predicting one direction and more about position sizing, diversification and the trade-off between risk and return.

Best “what went wrong?” answer

The biggest weakness is the granularity of the data. Annual data is useful for a long historical study, but it cannot capture intraday market reactions, exact macro surprises or trading costs. The next version would therefore move to daily total-return data and use walk-forward testing.

Best “why FICC & Equities?” connection

The project is interesting to me because it sits across several markets rather than treating equities in isolation. Rates affect valuations, FX reflects differences in monetary policy and growth, commodities can influence inflation, and credit tells you about risk appetite. FICC & Equities brings those relationships together in a real market environment.

15. Appendix: key formulas

Model	Formula / idea	Dummy explanation
Portfolio return	$R_p = \sum(w_i \times R_i)$	Weight each asset return and add the contributions.
Volatility	$\sigma_p = \sqrt{w^T \Sigma w}$	Combines each asset's volatility and how assets move together.
Sharpe ratio	$(R_p - R_f) / \sigma_p$	Return earned per unit of volatility above the cash/risk-free proxy.
Drawdown	Current value / previous peak - 1	Measures how far the portfolio has fallen from its previous high.

Beta	$\text{Cov}(R_{\text{asset}}, R_{\text{market}}) / \text{Var}(R_{\text{market}})$	Sensitivity to movements in the chosen benchmark.
Tracking error	$\text{StdDev}(R_{\text{strategy}} - R_{\text{benchmark}})$	How different the strategy's returns are from the benchmark.
Duration approximation	$\Delta P/P \approx -D \times \Delta y + 0.5C(\Delta y)^2$	Estimates bond price movement for a yield shock.
DV01	Dollar P&L for a 1bp yield move	A practical fixed-income risk measure.
Option delta	Change in option value for a small spot move	Measures directional sensitivity.
Option gamma	Change in delta for a small spot move	Measures how fast delta changes.
Option vega	Change in option value for a 1-point volatility change	Measures volatility sensitivity.
Option theta	Change in value for one day less to expiry	Measures time decay.
VaR	Chosen loss percentile of historical returns	A threshold estimate, not a guarantee.
Expected shortfall	Average loss beyond the VaR threshold	Gives more detail on the tail.
Monte Carlo	Repeat random paths using assumed return/volatility inputs	Shows a distribution of possible outcomes rather than one forecast.

Sources and important limitations

The workbook contains a Sources sheet with the underlying source URLs. The historical return series are attributed to Aswath Damodaran's historical return data; monetary-policy and current market inputs use Federal Reserve/FRED and Bank of England sources, with Brent linked to FRED/EIA. The project is educational research, not investment advice, and its results are historical rather than predictive.

Damodaran historical returns - S&P 500, Small Cap, bonds, real estate, gold, T-bills

https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html

FRED RIFSPFFNA - Fed Funds annual history

<https://fred.stlouisfed.org/series/RIFSPFFNA>

FRED / EIA Brent - Brent crude annual history

<https://fred.stlouisfed.org/series/DCOILBRENTU>

FRED SP500 - Current S&P snapshot

<https://fred.stlouisfed.org/series/SP500>

FRED DFF - Current Fed Funds

<https://fred.stlouisfed.org/series/DFF>

FRED DGS10 - Current US 10Y

<https://fred.stlouisfed.org/series/DGS10>

Bank of England - UK Bank Rate

<https://www.bankofengland.co.uk/boeapps/database/Bank-Rate.asp>

ExchangeRates.org.uk - GBP/USD reference

<https://www.exchangerates.org.uk/GBP-USD-spot-exchange-rates-history-2026.html>

Goldman Sachs FICC & Equities - Programme/role context

<https://www.goldmansachs.com/careers/our-firm/FICC-and-equities>

Goldman Sachs Degree Apprenticeships - Apprenticeship structure

<https://www.goldmansachs.com/careers/students/programs-and-internships/emea/degree-apprentices>